Yousung Lee

Graduate School of Mobility, KAIST

Personal Data

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Research Interests

- Vision-Language-Action Models
- Uncertainty Quantification and Failure Detection
- Safe and Trustworthy Robot Learning

Education

KAIST Daejeon, South Korea
M.S. Candidate in the Graduate School of Mobility (GPA: 4.0 / 4.3) Feb. 2025 – Present
Advisor: Prof. Dong Soo Har

Seoul National University of Science and Technology Seoul, South Korea
B.S. in Electronic Engineering (GPA: 4.11 / 4.5) Mar. 2021 – Feb. 2025

Publications

Perturbation-Based Epistemic Uncertainty for Failure Detection in Vision-Language-Action Models
Yousung Lee, Dongsoo Har
Under Review at a Robotics Conference
RSS Workshop on Trustworthy Embodied Foundation Models, 2026, Oral

Steering Sparse Autoencoder Latents to Control Dynamic Head Pruning in Vision Transformers (Student Abstract)
Yousung Lee, Dongsoo Har
AAAI Conference on Artificial Intelligence (Student Abstract & Poster), 2026, Poster

Vision-Based Tire Lifting for Parking by Dual Parking Robot
Neha Sengar, **Yousung Lee**, Kuk Won Ko, Dongsoo Har, Sangkeum Lee
IEEE Sensors Journal, 2026, Published

Research Experience

Real-World VLA Fine-Tuning and Failure Detection
Under Review at a Robotics Conference / RSS Workshop 2026 Oral
Integrated a TM5-900 manipulator with an AG95 gripper, fine-tuned a VLA policy on real-world demonstrations, and collected autonomous rollouts to develop and validate failure-detection methods under an object distribution shift.

Under-Vehicle Wheel Dataset Collection for Dual Parking Robots
IEEE Sensors Journal, 2026
Teleoperated a TurtleBot3 beneath vehicles to collect a partial-wheel image dataset used to develop and evaluate a vision-based tire-lifting system for dual parking robots.

Context-Aware Contrastive Value Maps for Zero-Shot Object Navigation
KAIST MO503: Introduction to Autonomous Vehicle, Final Project
Improved zero-shot ObjectNav by augmenting VLFM's semantic frontier scoring with BLIP-2-based context-aware contrastive prompts, evaluated on 2,000 HM3D episodes.